Elementary contraction of the thirty fixed-target entry panels

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This is a conditional consequence of the accepted object `INITIAL42_FIXED_TARGET_COARSE_V7`. It does not prove a count comparison, confinement to half amplitude, or the finite base. The numerical work is exact rational arithmetic on thirty probability intervals, with no Fourier-kernel execution.

Inputs and notation. Let X be a centered variance-one fixed-target maximizing witness at $n \geq N = 42$, with nonnegative excess over $c = C_{\mathrm{E}}$. The previously established entry supplies $nD \leq ND_I$, the moment ceilings in each closed probability panel $I = [l, u]$, and an actual central gap $(-A, B)$, where $A, B > 0$. Here $\rho = \mathbb{E}|X|^3$, $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$, $D = (3 + \sqrt{10})\rho - \gamma - 3J$, and J is the weighted gap sum. The final active rho rows also give $|X| \leq R = 157/50$: this is the maximum of their individually acquired support radii. All thirty nonempty panels, indexed 24 through 53, are retained initially. Their data are bound by SHA256 `61b7ad43e20046d7cbf7157e3feb61696331731e7be418f6a5fa535c2eee8bae`.

Write $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. The conditional means are $b = m/p$, $a = m/q$, and

$$\theta = 1 - \frac{m^2}{pq}, \qquad \ell = \frac{m}{pq} = \sqrt{\frac{1-\theta}{pq}}, \qquad \delta = 1 - \frac{A+B}{\ell}, \qquad \lambda = \frac{n\theta}{\ell^2}.$$

Thus δ is a count-coordinate inward displacement, not D. Let $V = b$ on the positive sign and $V = -a$ on the negative sign; write $W = X - V$. Define the two measures on nonnegative magnitudes by $\int f(x)d\mu_\pm(x) = \mathbb{E}[W^2\mathbf{1}_{\{\pm X>0\}}f(|X|)]$. Their masses sum to θ; the central gap ensures there is no atom at zero. Also $A \leq a$, $B \leq b$.

Retained analytic premise. We use the actual-law outer-gap inequality

$$D \geq (\sigma-1)\mathbb{E}X_+^3 + (\sigma+1)\mathbb{E}X_-^3 - 6mAB + 6BR_+ + 6AR_-, \quad \sigma = \sqrt{10}, \quad R_+ = \mathbb{E}X_+^2 - mB,\ R_- = \mathbb{E}X_-^2 - mA.$$

It is the input proved in the earlier weighted-gap analysis; arithmetic evaluation does not prove it. Substituting $\mathbb{E}X_+^3 = mb^2 + \int(x+2b)d\mu_+$ and $R_+ = \mu_+(\mathbb{R}) + m(b-B)$, and their negative-sign counterparts, gives

$$D \geq B_0 + E_0 + \int k_+ d\mu_+ + \int k_- d\mu_-, \tag{1}$$

where, with $p_* = (4 - \sqrt{10})/2$,

$$\begin{aligned} B_0 &= \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p-p_*)^2, \\ E_0 &= 6m\{(A+B)(a-A+b-B) + (a-A)(b-B)\}, \\ k_+(x) &= (\sigma-1)(x+2b) + 6B, \qquad k_-(x) = (\sigma+1)(x+2a) + 6A. \end{aligned}$$

The deterministic identity is $(\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2$. Every displayed cost is nonnegative.

A whole-panel iteration. Set $v_- = \min_{p\in I} pq$, $v_+ = \max_{p\in I} pq$. The maximum includes $1/4$ if I crosses $1/2$. Use $\sigma_l = 3.162277660 < \sigma < 3.162277661 = \sigma_u$. Initially $D \leq D_I < \sigma_l - 1$, and the gap bound $J \leq \rho + 2mAB$ gives

$$AB \geq z := (\sigma_l - 1 - D_I)/3 > 0.$$

Indeed $\rho \ge 1, \gamma \le \rho, m \le 1/2$ imply this inequality. With $a_u = \sqrt{u/(1-u)}$, $b_u = \sqrt{(1-l)/l}$, obtain

$$A \ge A_l = z/b_u, \quad B \ge B_l = z/a_u, \quad (A+B)/\ell \ge g = 2\sqrt{zv_-}, \quad \delta \le 1-g.$$

For a currently proved $\theta \le \Theta$, lower-bound the binary charge by

$$b_I = \frac{(6+2\sigma_l)(1-\Theta)^{3/2}}{\sqrt{v_+}} \operatorname{dist}(I, [(4-\sigma_u)/2, (4-\sigma_l)/2])^2.$$

All square roots are directed outward. If $r = D_I - b_I < 0$, the entire closed panel is impossible. Otherwise set

$$a_l = \sqrt{(1-\Theta)l/(1-l)}, \quad b_l = \sqrt{(1-\Theta)(1-u)/u},$$

$$K = \min\{(\sigma_l - 1)(B_l + 2b_l) + 6B_l,\ (\sigma_l + 1)(A_l + 2a_l) + 6A_l\} > 0.$$

Equation (1) gives $\theta \le r/K$. Intersect this bound with Θ, then recompute b_I, r. Put $m_l = \sqrt{(1-\Theta)v_-}$ and $\ell_l = \sqrt{(1-\Theta)/v_+}$. Since $E_0 \ge 6m\ell^2 g\delta$, we may update

$$\delta \le \min\{\delta_{\text{old}}, r/(6m_l\ell_l^2 g)\}, \quad g \ge 1-\delta, \quad A_l \leftarrow \max\{A_l, \ell_l(l-\delta)_+\}, \quad B_l \leftarrow \max\{B_l, \ell_l(1-u-\delta)_+\}.$$

Only previously proved bounds enter each update. Zero residual forces zero noise and displacement; no division by that residual is needed. The implementation performs eight rounds, rounding upper bounds upward and lower bounds downward to 10^{-10}.

Moment and weighted-noise exports. For every orientation, including $p > 1/2$,

$$1 - 2u - 2(1-u)\Theta \le \beta \le 1 - 2l + 2l\Theta. \tag{2}$$

This follows from $\beta = (1-\theta)(1-2p) + \mu_+(\mathbb{R}) - \mu_-(\mathbb{R})$. The formerly convenient upper replacement $1 - 2l + \Theta$ requires $l \le 1/2$ and must not be continued past that point.

For a coherent third-moment cap, put

$$L = R + 2\max(a_u, b_u), \quad k = \min\{\sigma_l - 1 + 6B_l/(R + 2b_u),\ \sigma_l + 1 + 6A_l/(R + 2a_u)\}.$$

The exact identity $\rho = \rho_B(p)(1-\theta)^{3/2} + \sum_{\pm} \int (x + 2a_{\pm}) d\mu_{\pm}$ gives a noise charge at most $\min(L\theta, r/k)$. Let $r_B = (1 - 2v_-)/\sqrt{v_-}$ and $s \le (3/2)\sqrt{1-\Theta}$. Integrating the derivative proves $(1-\theta)^{3/2} \le 1 - s\theta$ for $0 \le \theta \le \Theta$. The maximum of this affine, piecewise expression is therefore bounded by

$$\rho \le r_B + (L - sr_B)_+ \min\{\Theta, r/(kL)\}.$$

Intersect with the prior rho cap.

Finally (1) implies the joint inequality

$$D \ge b_I + 6m(A_l + B_l)\ell\delta + K\theta.$$

Multiplying by the actual n, using $nD \le ND_I$, $m\ell = 1 - \theta$, and $n\theta = (1-\theta)\lambda/(pq)$, proves, when $r > 0$,

$$\frac{\delta}{r/[6(1-\Theta)(A_l + B_l)]} + \frac{\lambda}{Nv_+ r/[K(1-\Theta)]} \le 1.$$

This retains the weighted resource; it is not obtained by multiplying an arbitrary unweighted noise cap by N.

Theorem 1 (Conditional all-panel result)**.** *The exact eight-round calculation excludes panels 24 and 53. On all 28 remaining panels,*

$$\frac{1}{4} \le p \le \frac{53}{100}, \quad \theta \le .0594425514, \quad \delta \le .0719539227, \quad \rho \le 1.4547288545, \quad \lambda \le .6314459997,$$

$$1.957063175 \le \ell \le 2.3094010768.$$

All endpoints and both signs are included.

Proof. The previous inequalities justify each recorded rational update. The input is the full accepted closed cover, and only a strictly negative residual excludes a panel. All 28 retained outputs and every update are in the exact log; the stated constants are their outward maxima or minima. No sampled parameter or favorable moment optimizer is used. □

A direct support-strip check and its limit. The retained fixed-target contact theorem, with sharp predecessor and nonnegative excess, bounds a support contact by $H_n(H, \pm t) + cQ(t)$, where $Q(t) = -t^3 + 3rt^2/2 + 3bt + (8/875)r$. Here r is a rho upper bound, and $b = \beta_u$ on the positive ray, $b = -\beta_l$ on the negative ray. For $t > \sqrt{3}$, set

$$A_t = (t^3 - 3t)/6, \quad H_4 = t^4 - 6t^2 + 3, \quad S = 81/80, \quad E_t = t^5/4100 + 3t/820 + 17(3t^2 - 2)/78720.$$

The heat expansion used in the preceding entry proof remains valid here: its fifth-derivative error has coefficient $3t^2 - 2$, since $t^2 - 1 > 0$. Using $\|\phi'''\|_\infty < 2\phi(0)$, $\phi(0) < 2/5$ and $\sqrt{n-1} > 6$, without any sign assumption on H_4, yields

$$H_n(H, \pm t) \le \tfrac{2}{5} S^3 (A_t + |H_4|/72) + S^3 E_t.$$

Thus a negative boundary value of this expression plus $(819/2000)Q(t)$ excludes that boundary, provided $Q(t) < 0$. The exact derivative estimate is $|\partial_t H_n| < \frac{2}{5}(t^2/2 + 1/2 + 8/875)$. Check $Q, Q', Q'' < 0$ and

$$L = \tfrac{2}{5}(t^2/2 + 1/2 + 8/875) + (819/2000)Q', \quad L', L'' < 0.$$

Their higher derivatives are negative constants, so both rays are excluded beyond the boundary, for every raw threshold H. These Gaussian/contact premises are retained analytical inputs; none is proved by a numerical PASS.

Choose the two boundaries $t_- = \ell_l(l + 3/5)$ and $t_+ = \ell_l(1 - u + 3/5)$. All satisfy $t_\pm > \sqrt{3}$. Both pass on panels 40–52, proving $Y = X/\ell + p \in (-3/5, 8/5)$ there. On panels 25–39 the negative ray passes but the positive boundary upper estimate is nonnegative. This is failure of that sufficient estimate, not a counterexample to the strip theorem. No strip claim is made on those fifteen panels, and no half-amplitude confinement is claimed anywhere.

Adversarial examples. The standardized three-point law with atoms and masses

$$(-26/25, 12/25), \quad (1/2, 52/6977), \quad (7226/7475, 89401/174425)$$

has $p = 13/25, \theta = 1/625, \beta = -599/15625$. Its beta exceeds $1 - 2p + \theta = -24/625$ by $1/15625$. It verifies the general formula (2) and refutes the half-interval shortcut outside its hypothesis; it is not a counterexample to Berry–Esseen.

A standardized Bernoulli law at $p = 13/50$ has $\theta = \delta = 0$, $\beta = 12/25$,

$$\rho = \frac{769}{25\sqrt{481}}, \qquad D = \frac{769\sqrt{10} - 2043}{25\sqrt{481}}.$$

It satisfies the remaining moment/defect constraints of panel 25. Hence these necessary constraints alone cannot exclude that whole panel. This law is not asserted to be a nonnegative-excess maximizing witness. The checker also verifies the gap-cost identity on symmetric and skew Bernoulli laws, their reflection, a four-point law and small distant atoms. Seven invalid controls reject. All work remains separate from the finite base.